

Simulations

Loïc Pages

2025-11-06

Introduction

```
rm(list=ls())  
library(knitr)  
library(spaMM)
```

```
## Registered S3 methods overwritten by 'registry':  
##   method          from  
##   print.registry_field proxy  
##   print.registry_entry proxy
```

```
## spaMM (Rousset & Ferdy, 2014, version 4.6.3) is loaded.  
## Type 'help(spaMM)' for a short introduction,  
## 'news(package='spaMM')' for news,  
## and 'citation('spaMM')' for proper citation.  
## Further infos, slides, etc. at https://gitlab.mbb.univ-montp2.fr/francois/spamm-ref.
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.1      v tibble     3.2.1  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(popbio)
```

```
setwd("/media/loic/Commun/OTravail/2025 IPM Centauree ISEM/2025 Stage ISEM/Code")
```

```
spaMM.options(separation_max=70)  
MatrixDim <- 50
```

```

load("IPM/ModelsAIC")
#Flowering function
flr0 <- function(x, a, Pop, year) {
  fake_data <- expand.grid(
    year = year,
    Pop = Pop,
    Size0Mars = unique(x),
    Age = a)
  mu <- plogis(predict(Flowglm1, newdata = fake_data, allow.new.levels = T, type = "link"))
  mu2 <- aggregate(mu, list(fake_data$Size0Mars), mean)
  return(mu2$V1)
}

minsize <- 0.5
maxsize <- 25
maxsize1 <- 15
n.size <- 50
h <- (maxsize - minsize) / n.size
h1 <- (maxsize1 - minsize) / n.size
b = minsize + c(0:n.size) * h

#sizes for an individual of age a
#midpoint
ymid.a = 0.5 * (b[1:n.size] + b[2:(n.size + 1)])
#intervals
yI.a <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.a[i,1] <- b[i]
  yI.a[i,2] <- b[i+1]
}

b1 = minsize + c(0:n.size) * h1
#sizes for an individual of age 1
#midpoint
ymid.1 = 0.5 * (b1[1:n.size] + b1[2:(n.size + 1)])
#intervals
yI.1 <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.1[i,1] <- b1[i]
  yI.1[i,2] <- b1[i+1]
}

```

```

population <- c("E2", "E1", "Au", "Po", "Pe", "Cr")
annees <- 1995:2022

```

```

Nf_obs <- c(494, 420, 604, 638, 343, 490, 631, 482, 815, 435, 613, 794, 210, NA, 322, 216, 277, 386, 565, 254, 369, 407, 148,
Nf <- read.csv("IPM/flower_data.csv")
Nf <- type_convert(Nf, cols(
  Auzils = col_double(),
  Cruzades = col_double(),
  Enferrets.1 = col_double(),
  Enferrets.2 = col_double(),

```

```

    Peyrals = col_double(),
    Portes = col_double()
  ))
names(Nf) <- c("year", "E2", "E1", "Au", "Po", "Pe", "Cr")
Nf <- Nf %>% filter(year!=1994, year!=2025, year!=2024, year!=2023)

```

```

load("IPM/IPMKernalPopYear_AIC")
MatrixIPM <- IPMKernalPopYear_AIC
ME2 <- list()
ME1 <- list()
MAu <- list()
MPo <- list()
MPe <- list()
MCr <- list()

for (i in 1:28){
  ME2[[i]] <- MatrixIPM[[i]]
  ME1[[i]] <- MatrixIPM[[i+28]]
  MAu[[i]] <- MatrixIPM[[i+28*2]]
  MPo[[i]] <- MatrixIPM[[i+28*3]]
  MPe[[i]] <- MatrixIPM[[i+28*4]]
  MCr[[i]] <- MatrixIPM[[i+28*5]]
}

```

Simulations de tailles de population pendant 50 ans

```

#p: 1>Enferet2, 2>Enferet1, 3>Auzils, 4>Portes, 5>Peyrals, 6>Cruzades
simu <- function(p, nbroyears = 50){
  N <- NULL
  Nia <- array(data = 0, dim = c(nbroyears, 8, MatrixDim))

  Pop <- population[p]
  if (Pop=="E2") {Mat <- ME2}
  if (Pop=="E1") {Mat <- ME1}
  if (Pop=="Au") {Mat <- MAu}
  if (Pop=="Po") {Mat <- MPo}
  if (Pop=="Pe") {Mat <- MPe}
  if (Pop=="Cr") {Mat <- MCr}

  # Première année
  year <- annees[28]

  # Définition de la matrice et du vecteur propre
  M <- Mat[[28]]
  A <- eigen.analysis(M)
  W <- function(age) {
    return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])
  }

  # Densité estimée d'individus fleuris
  Nf_dens <- sum(W(1) * flr0(ymid.1, 1, Pop, year))
  for (a in 2:8) {
    Nf_dens <- Nf_dens + sum(W(a) * flr0(ymid.a, a, Pop, year))
  }
}

```

```

}
# Nombre d'individus total dans la population Pop la première année, utilisant les données observées
N[1] <- Nf[1, p + 1] / Nf_dens
# Nombre d'individus dans chaque classe de taille et d'age
for (a in 1:8) {
  for (i in 1:MatrixDim) {
    Nia[1, a, ] <- N[1] * W(a)
  }
}

# Itération sur les années suivantes
for (yr in 2:nbrofyears) {
  year <- annees[yr]
  # Définition de la matrice et du vecteur propre
  M <- sample(Mat)[[1]]
  A <- eigen.analysis(M)
  W <- function(age) {
    return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])
  }
  Fec <- function(a) {
    #a age de plante mère
    return(M[1:50, ((a - 1) * 50 + 1):(a * 50)])
  }
  P <- function(a) {
    #a age de plante à t
    if (a == 8) {
      return(M[((50*(a-1))+1):(50*a), (50*(a-1)+1):(50*a)])
    }
    return(M[((50*a)+1):(50*(a+1)), (50*(a-1)+1):(50*a)])
  }

  # Calcul du nombre d'individus dans chaque classe de taille et d'age en utilisant le kernel
  for (a in 1:8) { # Nouvelles plantules produites
    Nia[yr, 1, ] <- Nia[yr, 1, ] + Fec(a) %*% Nia[yr - 1, a, ]
  }
  for (a in 2:8) { # Croissance-survie des plantes
    Nia[yr, a, ] <- P(a) %*% Nia[yr - 1, a - 1, ]
  }
  Nia[yr, 8, ] <- Nia[yr, 8, ] + P(8) %*% Nia[yr - 1, 8, ]

  N[yr] <- sum(Nia[yr,,])
}
return(N)
}

```

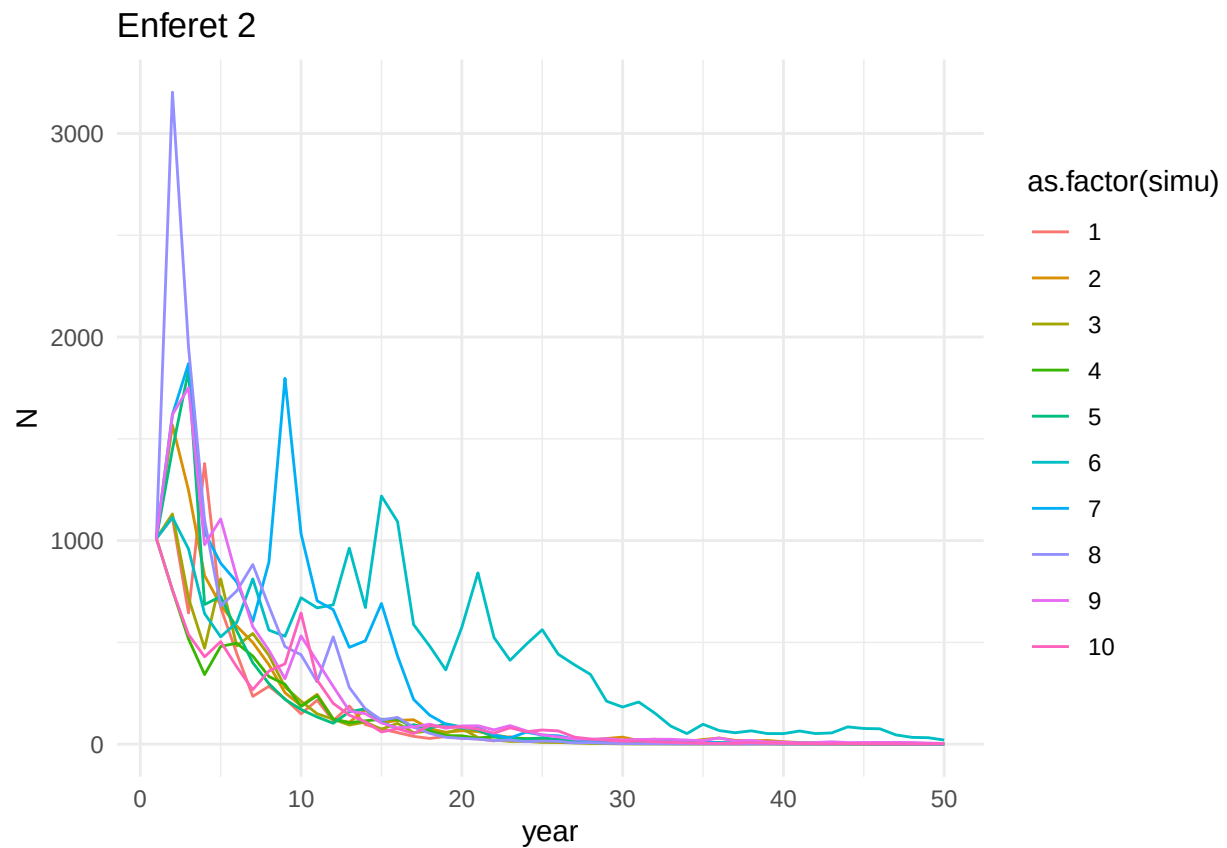
```

nbsimu <- 10
simuE2 <- expand.grid(
  year = 1:50,
  simu=1:nbsimu)
for(i in 1:nbsimu){
  simuE2$N[simuE2$simu==i] <- simu(p=1)
}

ggplot(simuE2, aes(x = year, y = N, color = as.factor(simu))) +

```

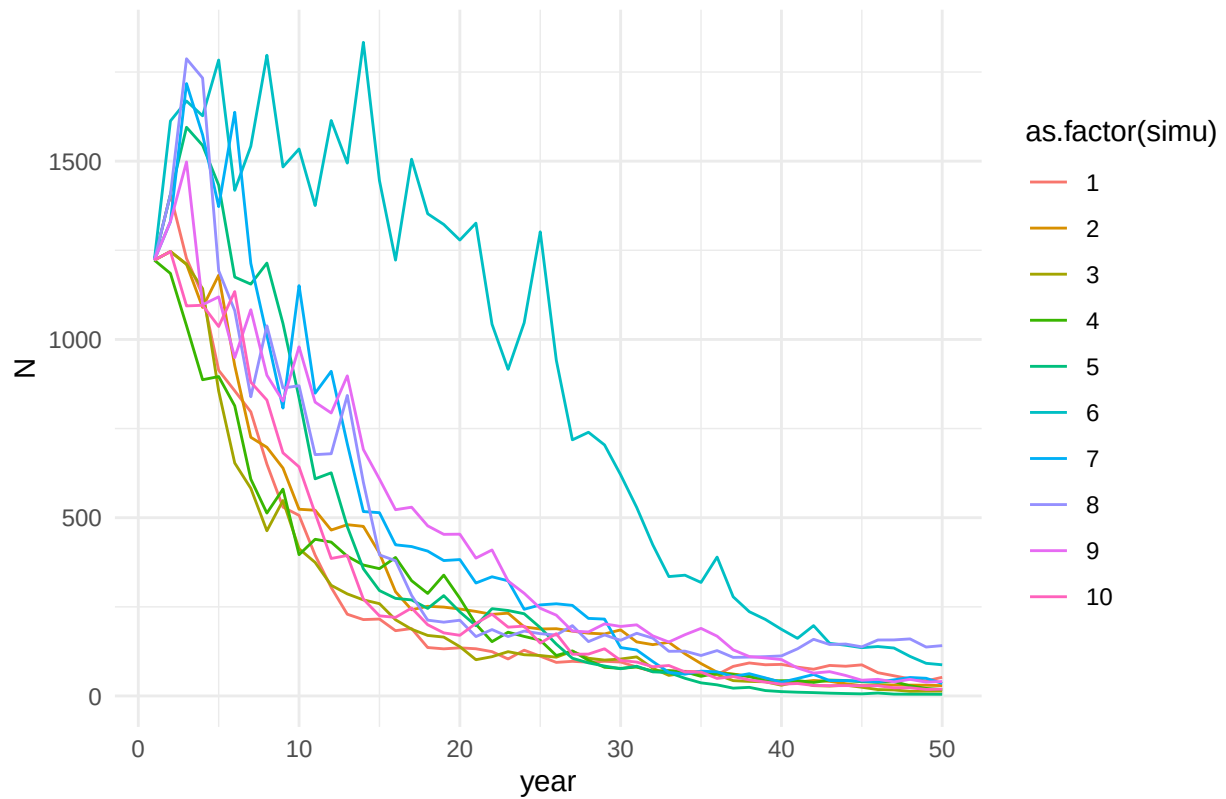
```
geom_line() +
theme_minimal()+
labs(title="Enferet 2")
```



```
nbsimu <- 10
simuE1 <- expand.grid(
  year = 1:50,
  simu=1:nbsimu)
for(i in 1:nbsimu){
  simuE1$N[simuE1$simu==i] <- simu(p=2)
}

ggplot(simuE1, aes(x = year, y = N, color = as.factor(simu))) +
  geom_line() +
  theme_minimal()+
  labs(title="Enferet 1")
```

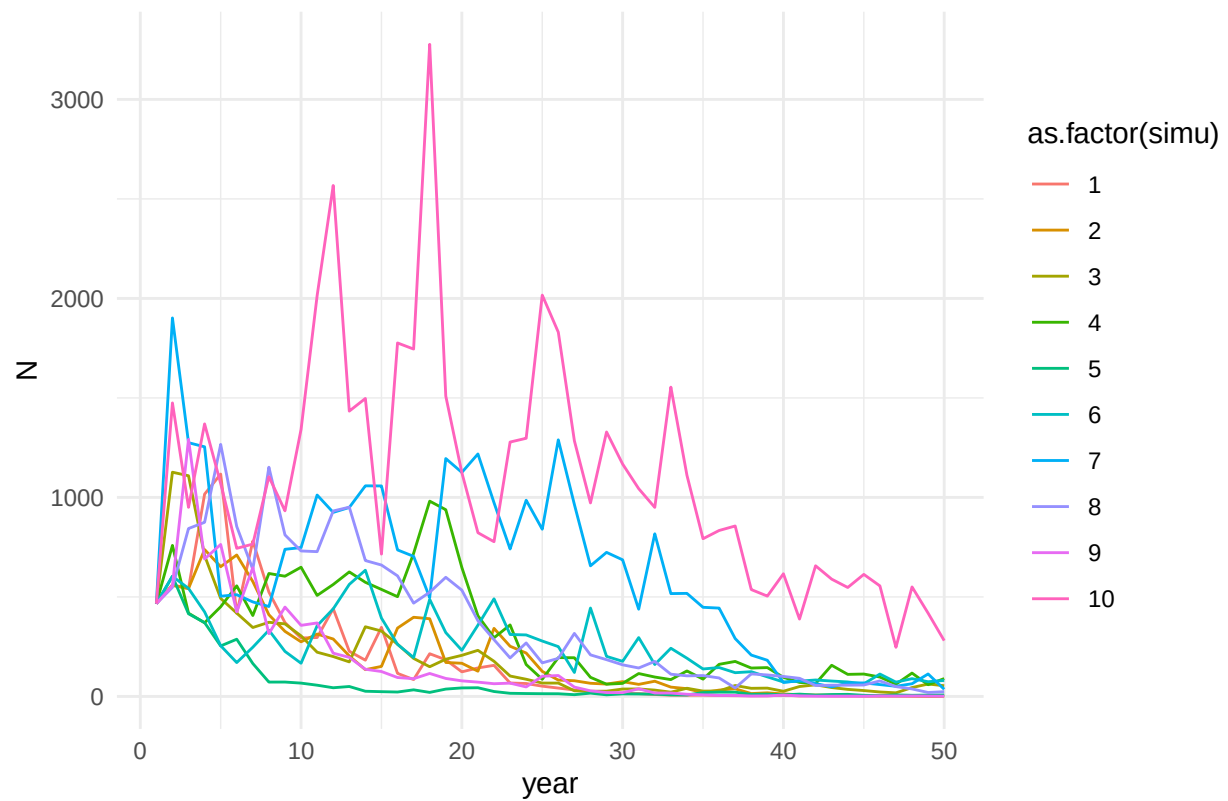
Enferet 1



```
nbsimu <- 10
simuAu <- expand.grid(
  year = 1:50,
  simu=1:nbsimu)
for(i in 1:nbsimu){
  simuAu$N[simuAu$simu==i] <- simu(p=3)
}

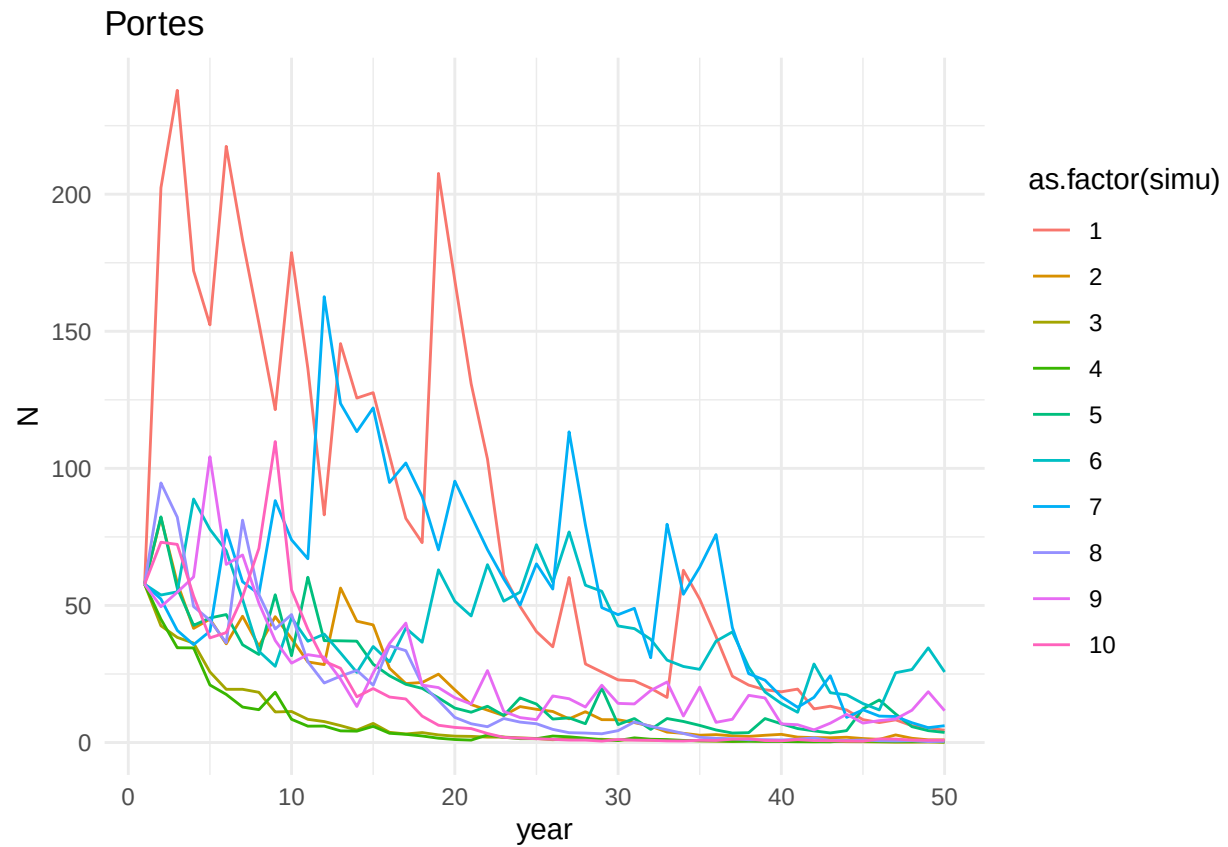
ggplot(simuAu, aes(x = year, y = N, color = as.factor(simu))) +
  geom_line() +
  theme_minimal()+
  labs(title="Auzils")
```

Auzils



```
nbsimu <- 10
simuPo <- expand.grid(
  year = 1:50,
  simu=1:nbsimu)
for(i in 1:nbsimu){
  simuPo$N[simuPo$simu==i] <- simu(p=4)
}

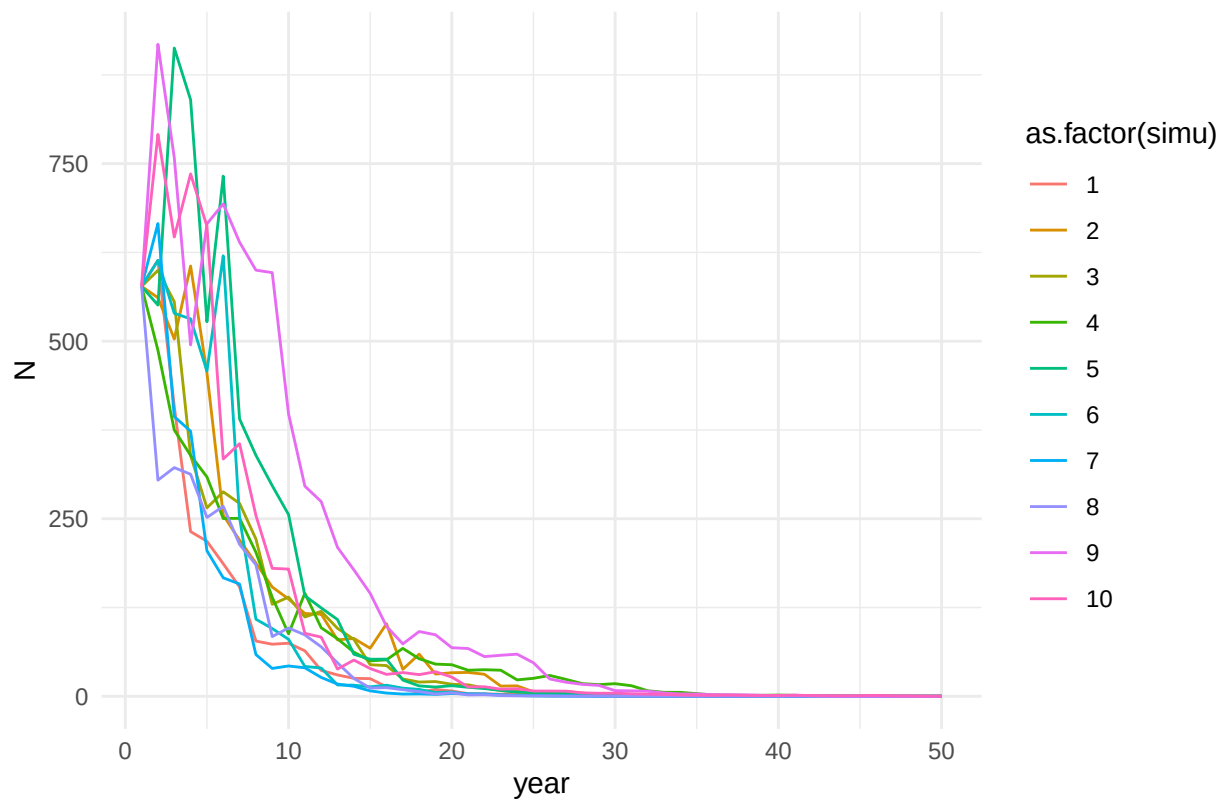
ggplot(simuPo, aes(x = year, y = N, color = as.factor(simu))) +
  geom_line() +
  theme_minimal()+
  labs(title="Portes")
```



```
nbsimu <- 10
simuPe <- expand.grid(
  year = 1:50,
  simu=1:nbsimu)
for(i in 1:nbsimu){
  simuPe$N[simuPe$simu==i] <- simu(p=5)
}

ggplot(simuPe, aes(x = year, y = N, color = as.factor(simu))) +
  geom_line() +
  theme_minimal()+
  labs(title="Peyrals")
```

Peyrals



```
nbsimu <- 10
simuCr <- expand.grid(
  year = 1:50,
  simu=1:nbsimu)
for(i in 1:nbsimu){
  simuCr$N[simuCr$simu==i] <- simu(p=6)
}

ggplot(simuCr, aes(x = year, y = N, color = as.factor(simu))) +
  geom_line() +
  theme_minimal()+
  labs(title="Cruzade")
```

